

Aim: Back Propagation.**1.Importing the Main libraries.**

```
%matplotlib inline
import matplotlib.pyplot as plt
import matplotlib.animation as animation
from IPython.display import HTML
```

Matplotlib is building the font cache; this may take a moment.

2.Setting the co-ordinates of the Nodes.

```
#NODE COORDINATES
input_nodes = [(0,1),(0,-1)]
hidden_nodes = [(2,1.5),(2,0),(2,-1.5)]
output_node = [(4,0)]
```

3.Code and output for forward and backward propagation.

```
fig, ax = plt.subplots(figsize = (8,5))
def draw_network(frame):
    ax.clear()
    ax.set_xlim(-1,5)
    ax.set_ylim(-3,3)
    ax.axis('off')

    for x, y in input_nodes:
        ax.plot(x, y, 'o', markersize=20, color='red')
    for x, y in hidden_nodes:
        ax.plot(x, y, 'o', markersize=20, color='yellow')
    for x, y in output_node:
        ax.plot(x, y, 'o', markersize=20, color='green')

    for ix, iy in input_nodes:
        for hx, hy in hidden_nodes:
            ax.annotate('', xy = (hx-0.2,hy), xytext = (ix+0.2, iy),
                        arrowprops = dict(arrowstyle = '->', color = 'gray', lw = 2 if frame < 15 else 1))

    for hx, hy in hidden_nodes:
        for ox, oy in output_node:
            ax.annotate('', xy = (ox-0.2, oy), xytext = (hx+0.2, hy),
                        arrowprops = dict(arrowstyle = '->', color = 'gray', lw = 2 if frame < 15 else 1))
```

```

if frame >= 15:
    for ox, oy in output_node:
        for hx, hy in hidden_nodes:
            ax.annotate('', xy = (hx+0.2, hy), xytext = (ox-0.2, oy),
                        arrowprops = dict(arrowstyle = '->', color = 'red', lw = 2))

    for hx, hy in hidden_nodes:
        for ix, iy in input_nodes:
            ax.annotate('', xy = (ix+0.2, iy), xytext = (hx-0.2, hy),
                        arrowprops = dict(arrowstyle = '->', color = 'red', lw = 2))

ax.text(-0.6, 1.2, 'Input 1', fontsize = 10)
ax.text(-0.6, -0.8, 'Input 2', fontsize = 10)
ax.text(4.3, 0, 'Output', fontsize = 10)

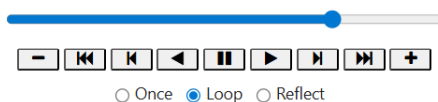
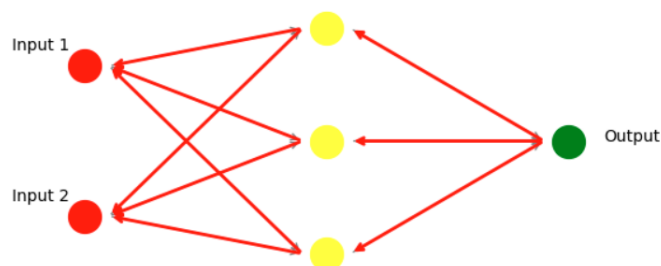
if frame < 15:
    ax.set_title("Forward Pass: Signals ->", fontsize = 14)
else:
    ax.set_title("Backpropagation: Errors <-", fontsize = 14, color = 'red')

#HTML
ani = animation.FuncAnimation(fig, draw_network, frames = 30, interval = 300, repeat = True)
HTML(ani.to_jshtml())

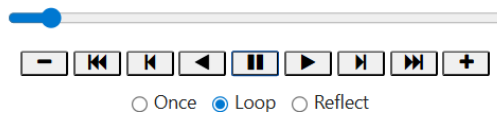
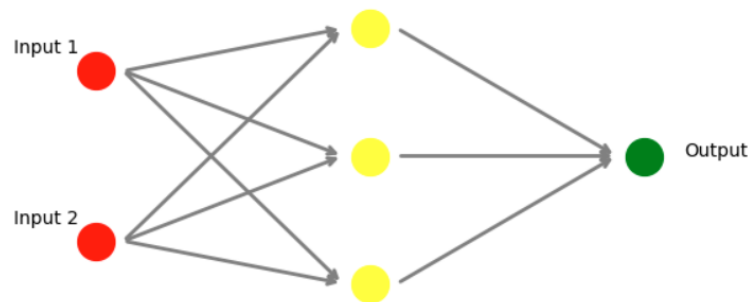
```

OUTPUT:

Backpropagation: Errors <-



Forward Pass: Signals ->



Forward Pass: Signals ->

